NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Marks: 20	**QUIZ-3**	**Time: 20 Minutes**

Subject: Probability and Statistics **Section: D**
Instructor: Hazber Samson Date: 17-03-2026
Student Name_____ Roll No_____

Q1. **(4+4+4+4+4=20)**

If the joint probability distribution of *X* and *Y* is given by

$$f(x,y)=\frac{x+y}{30} \quad for\ x=0,1,2,3 : y=0,1,2$$

(a) Construct joint probability distribution and then evaluate the marginal distributions of *X* and Y.
(b) Find *P*(*Y* = 2 | *X* = 1) and $P(X+Y\le 3)$
(c) Check whether X and Y are independent or not.
(d) Find $E(XY), E(X), E(Y)$
(e) Find $COV(X,Y)$ *and* $Corr(X,Y)$

GOOD LUCK !